



PROCEDURE:

Make the Connections as shown in figure.

(a) Initial settings: Keep the dc machine (separately excited) field rheostat at maximum resistance position. The TPST, DPST and SPST switches are kept open. The rotor rheostat should be kept at maximum position.

(b) Execution of the Experiment: Switch on; tie three phase supply by closing the TPST switch. Induction machine will start as induction motor. As the motor gathers speed, gradually cut off the rotor resistance. Now, switch on the DC supply by closing the DPST switch. **Decrease the** resistance of the field rheostat (i.e., excitation is increased) and **note the** voltmeter reading across the SPST switch. If it is increasing, switch off the dc supply **and** interchange the armature terminals A & AA. Again switch on the dc supply **keeping the** motor field rheostat at maximum position. Decrease the resistance of the field rheostat gradually till the voltmeter reading across the SPST switch zero. Now, the SPST switch is closed.

Again increase the excitation till the dc ammeter shows rated current (10A). wattmeter should be such that the wattmeter reading is positive. Note down **all the meter readings** and take 4 or 10 oscillations. Now, **decrease the excitation in steps for different** values of dc ammeter and note all the readings each time. This procedure is continued till the wattmeter reading is zero. Note the speed using tachometer.

If the excitation is again decrease, induction machine will be in generator mode. Now, interchange the connections of the pressure coil of the wattmeter **using the** reversing switch, Decrease the excitation and take readings for different values of dc ammeter. This procedure is continued till the dc ammeter reading reaches rated value. wattmeter should be such that the wattmeter reading is positive. Note down **all the meter readings** and take 4 or 10 oscillations. Now, **decrease the excitation in steps for different** values of dc ammeter and note all the readings each time. This procedure is continued till the wattmeter reading is zero. Note the speed using tachometer.

If the excitation is again decrease, induction machine will be in generator mode. Now, interchange the connections of the pressure coil of the wattmeter **using the** reversing switch, Decrease the excitation and take readings for different values of dc ammeter. This procedure is continued till the dc ammeter reading reaches rated value. **Before switching off the dc supply, increase the excitation till the dc ammeter reads zero. Now switch off the ac supply also.**

Result

- Performance characteristics were plotted while the induction machine is operating as generator and motor.
- Hysteresis power = _____ W